ICT Level 2

Lab Manual & Student Workbook

Level 2 - Class 7

Level: Level 2

Class: Class 7

Activities: 24 Station Slots (12 Activities x 2 Cohorts) + Dormitory Tasks

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How to Run This Lab - 25/25 Station Rotation

Class of 50 splits into two fixed cohorts of 25. While Cohort A works at the 10 computers in triads/dyads, Cohort B does the matching paper-pack task in the library. Cohorts swap next lesson, so every student gets hands-on time every alternate session.

Roles (laminated cards) - rotate every 7 minutes so all three students drive within one session:

Lab Roles (25/25 Station - 10 systems, 50 students: 5 as 3+2 Hybrid)

Driver	Navigator	Checker	Scribe	Timekeeper
At keyboard	Reads steps	Tests	Writes	Presents

40-minute timeline

- 0-3 min: Login, collect Group Folder and Role Cards
- 3-8 min: Whole-class retrieval (3 questions) plus teacher demo on projector
- 8-28 min: Triad work on the activity below (rotate Driver every 7 min); Library cohort does the paper pack
- 28-32 min: Two groups present 60 seconds each; peers give one improvement suggestion
- 32-36 min: Save to shared drive and screenshot result
- 36-40 min: Individual exit ticket (2 questions) - collected at the door

Noise levels: 0 silent, 1 whisper within triad, 2 presenting. Librarian supervises the library station using the printed paper packs.

Chapter 1: Networking and Internet

Activity 1.1: Network Configuration Check

Activity 1.1: Network Configuration Check

Duration: 40 minutes

Resources Needed:

- Computer with internet access
- Windows Control Panel
- Network and Sharing Centre
- Printed network diagram worksheet

Procedure: 1. Linked lesson: Level 2 Ch 1 2. (Driver) Open Control Panel and navigate to Network and Sharing Centre. 3. (Navigator) Click on 'Change adapter settings' to view available network connections. 4. (Checker) Right-click on the active network adapter and select 'Status'. 5. (Driver) Click 'Details' and note down the IPv4 Address, Subnet Mask, and Default Gateway. 6. (Navigator) Open Command Prompt and type `ipconfig /all` to view full configuration. 7. (Checker) Record the DNS Server addresses from the command output. 8. (Driver) Compare the information from GUI and Command Prompt. 9. (Navigator) Draw a simple network diagram showing your computer's position in the network. 10. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between IPv4 and IPv6 addresses? 3. Why does a computer need a Default Gateway? 4. What is the role of DNS Server in network communication? 5. How is a static IP address different from a dynamic IP address?

Activity 1.2: Email Setup and Practice

Activity 1.2: Email Setup and Practice

Duration: 40 minutes

Resources Needed:
- Computer with internet access
- Web browser
- Email service account (Gmail or Outlook)
- Printed email etiquette guide

Procedure: 1. Linked lesson: Level 2 Ch 1 2. (Driver) Open a web browser and navigate to mail.google.com or outlook.live.com. 3. (Navigator) Sign in with your school email credentials. 4. (Checker) Compose a new email with a proper subject line and greeting. 5. (Driver) Add at least two recipients using CC and one using BCC. 6. (Navigator) Attach a text file and a image file to the email. 7. (Checker) Draft a formal email to your teacher requesting permission for a lab session. 8. (Driver) Check the Sent folder to verify your email was sent successfully. 9. (Navigator) Reply to a received email and forward another email to a classmate. 10. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between CC and BCC in email? 3. Why should you not share your email password with others? 4. What are the components of a professional email address? 5. How can you recall a sent email in Gmail?

Chapter 2: Advanced Graphics

Activity 2.1: GIMP Interface Exploration

Activity 2.1: GIMP Interface Exploration

Duration: 40 minutes

Resources Needed:
- Computer with GIMP installed
- Sample image files (JPG, PNG)
- GIMP quick reference guide
- Printed interface layout worksheet

Procedure: 1. Linked lesson: Level 2 Ch 2 2. (Driver) Launch GIMP from the Start Menu or desktop shortcut. 3. (Navigator) Identify the Toolbox, Menu Bar, Tool Options, and Layer Dialog. 4. (Checker) Open a sample image using File > Open. 5. (Driver) Explore the Toolbox tools: Selection, Paint, Transform, and Text. 6. (Navigator) Use the Zoom tool to examine image details at different levels. 7. (Checker) Open the Image Properties dialog and note the image dimensions and resolution. 8. (Driver) Save the image in three different formats: XCF, PNG, and JPEG. 9. (Navigator) Experiment with the Brightness/Contrast adjustment under Colors menu. 10. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between XCF and PNG file formats? 3. Why is the layer system important in image editing? 4. What is the purpose of the Undo History dialog? 5. How does changing image resolution affect file size and quality?

Activity 2.2: Layer-based Editing

Activity 2.2: Layer-based Editing

Duration: 40 minutes

Resources Needed:
- Computer with GIMP installed
- Background image file
- Overlay text image files
- Layer transparency reference sheet

Procedure: 1. Linked lesson: Level 2 Ch 2 2. (Driver) Open GIMP and create a new image with dimensions 1024x768 pixels. 3. (Navigator) Add a background colour fill to the first layer. 4. (Checker) Create a new transparent layer using Layer > New Layer. 5. (Driver) Use the Text tool to add text on the new layer. 6. (Navigator) Adjust the text layer opacity to 50% and observe the effect. 7. (Checker) Add another layer and draw a shape using the Selection and Fill tools. 8. (Driver) Experiment with layer blending modes: Multiply, Screen, Overlay. 9. (Navigator) Merge all visible layers and export the final image as PNG. 10. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What happens to hidden layers when you export an image as JPEG? 3. How do blending modes affect the appearance of overlapping layers? 4. Why is it recommended to work with layers instead of a flat image? 5. How can you rearrange the stacking order of layers?

Chapter 3: Multimedia

Activity 3.1: Audio Mixing with Audacity

Activity 3.1: Audio Mixing with Audacity

Duration: 40 minutes

Resources Needed:
- Computer with Audacity installed
- Two or more audio files (WAV or MP3)
- Headphones
- Audio mixing tutorial handout

Procedure: 1. Linked lesson: Level 2 Ch 3 2. (Driver) Launch Audacity from the Start Menu. 3. (Navigator) Import the first audio file using File > Import > Audio. 4. (Checker) Import a second audio file; it will appear as a new track. 5. (Driver) Use the selection tool to highlight a section of the first track. 6. (Navigator) Apply

Fade In effect to the beginning of the first track. 7. (Checker) Apply Fade Out effect to the end of the second track. 8. (Driver) Adjust the gain slider on each track to balance the volume levels. 9. (Navigator) Use the Time Shift tool to align the tracks for proper mixing. 10. (Checker) Play back the mixed audio and make final adjustments. 11. (Driver) Export the final mix as a WAV file using File > Export Audio. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the purpose of the Gain slider on each track? 3. How does the Time Shift tool differ from the Selection tool? 4. Why is it important to preview the mix before exporting? 5. What audio format is best for sharing online and why?

Activity 3.2: Presentation Creation

Activity 3.2: Presentation Creation

Duration: 40 minutes

Resources Needed:
- Computer with LibreOffice Impress or PowerPoint
- Sample presentation templates
- Clip art and image files
- Presentation design checklist

Procedure: 1. Linked lesson: Level 2 Ch 3 2. (Driver) Open LibreOffice Impress and select a blank presentation. 3. (Navigator) Choose a design template and apply it to all slides. 4. (Checker) Create a title slide with the presentation title and your name. 5. (Driver) Add at least five content slides with headings and bullet points. 6. (Navigator) Insert at least two images and one chart into your presentation. 7. (Checker) Add slide transitions using the Slide Transition panel. 8. (Driver) Insert speaker notes for at least two slides. 9. (Navigator) Use the Slide Sorter view to rearrange slides. 10. (Checker) Run the slideshow in Presenter Mode and practice delivery. 11. (Driver) Save the presentation in ODP and export a copy as PDF. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the purpose of Speaker Notes in a presentation? 3. Why should you use consistent fonts and colours throughout? 4. How does the Slide Sorter view help in organising a presentation? 5. What are the advantages of exporting a presentation as PDF?

Chapter 4: Data Handling

Activity 4.1: Spreadsheet Formulas Practice

Activity 4.1: Spreadsheet Formulas Practice

Duration: 40 minutes

Resources Needed:
- Computer with LibreOffice Calc or Excel

- Sample student marks dataset
- Formula reference sheet
- Printed practice worksheet

Procedure: 1. Linked lesson: Level 2 Ch 4 2. (Driver) Open LibreOffice Calc and create a new spreadsheet. 3. (Navigator) Enter column headers: Name, Subject1, Subject2, Subject3, Total, Average. 4. (Checker) Enter at least 10 rows of student data with marks in three subjects. 5. (Driver) Use the SUM formula to calculate total marks for the first student. 6. (Navigator) Use the AVERAGE formula to calculate the average marks. 7. (Checker) Copy the formulas down for all students using the fill handle. 8. (Driver) Use the MAX and MIN functions to find highest and lowest marks. 9. (Navigator) Apply conditional formatting to highlight marks below 40. 10. (Checker) Format the cells to display values with appropriate decimal places. 11. (Driver) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Navigator) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between a formula and a function in a spreadsheet? 3. How does absolute cell referencing (A1) differ from relative referencing (A1)? 4. Why is conditional formatting useful for data analysis? 5. What happens if you delete a cell that is referenced by a formula?

Activity 4.2: Chart Creation

Activity 4.2: Chart Creation

Duration: 40 minutes

Resources Needed:

- Computer with LibreOffice Calc or Excel
- Spreadsheet from Activity 4.1
- Chart types reference guide
- Printed chart evaluation worksheet

Procedure: 1. Linked lesson: Level 2 Ch 4 2. (Driver) Open the spreadsheet created in Activity 4.1. 3. (Navigator) Select the data range containing Names and Total marks. 4. (Checker) Insert a Bar Chart using Insert > Chart. 5. (Driver) Customise the chart title, axis labels, and legend position. 6. (Navigator) Change the chart type to Column Chart and observe the difference. 7. (Checker) Create a Pie Chart showing the distribution of total marks. 8. (Driver) Add data labels to the pie chart to display percentages. 9. (Navigator) Create a Line Chart showing trends across subjects for each student. 10. (Checker) Move the charts to a separate sheet for better visibility. 11. (Driver) Export the chart as an image file for use in a report. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. When would you use a Bar Chart versus a Pie Chart? 3. Why is it important to label chart axes clearly? 4. How does a Line Chart help in identifying trends? 5. What factors should you consider when choosing a chart type?

Chapter 5: Programming with Scratch

Activity 5.1: Conditional Logic Game

Activity 5.1: Conditional Logic Game

Duration: 40 minutes

Resources Needed:

- Computer with Scratch (scratch.mit.edu)
- Conditional logic flowchart handout
- Game design template
- Sample Scratch projects for reference

Procedure: 1. Linked lesson: Level 2 Ch 5 2. (Driver) Open Scratch and start a new project. 3. (Navigator) Choose a sprite as the main character and a backdrop. 4. (Checker) Add an event block: When green flag clicked. 5. (Driver) Use a Forever loop with If-Then-Else blocks for movement. 6. (Navigator) Program the sprite to move left and right using arrow keys. 7. (Checker) Add a second sprite as an obstacle with collision detection. 8. (Driver) Use the 'touching' sensing block to detect collisions. 9. (Navigator) Add a score variable that increases when avoiding obstacles. 10. (Checker) Use a 'say' block to display 'Game Over' when the sprite touches the obstacle. 11. (Driver) Test the game and adjust speed and difficulty settings. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between If-Then and If-Then-Else blocks? 3. Why is the Forever loop essential for a game? 4. How would you add a timer to your game? 5. What happens if you remove the Forever loop from your script?

Activity 5.2: Sprite Interaction Project

Activity 5.2: Sprite Interaction Project

Duration: 40 minutes

Resources Needed:

- Computer with Scratch (scratch.mit.edu)
- Sprite interaction examples
- Message and broadcast reference sheet
- Project planning worksheet

Procedure: 1. Linked lesson: Level 2 Ch 5 2. (Driver) Open Scratch and create a new project with at least three sprites. 3. (Navigator) Design a short story where sprites interact with each other. 4. (Checker) Use the 'broadcast' block to send messages between sprites. 5. (Driver) Program Sprite1 to say something and broadcast a message. 6. (Navigator) Program Sprite2 to receive the message and respond. 7. (Checker) Add costume changes to create simple animations. 8. (Driver) Use the 'switch costume' block within a loop for animation. 9. (Navigator) Program Sprite3 to move towards Sprite1 when a message is received. 10. (Checker) Add sound effects using the 'play sound' block. 11. (Driver) Test the complete interaction sequence and refine timing. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the advantage of using broadcast messages over direct sprite control? 3. How can you create a conversation between two sprites? 4. What role do costumes play in sprite animation? 5. How would you synchronise actions between multiple sprites?

Chapter 6: Digital Citizenship

Activity 6.1: Password Strength Testing

Activity 6.1: Password Strength Testing

Duration: 40 minutes

Resources Needed:

- Computer with internet access
- Password strength checker website
- Password security guidelines handout
- Printed password creation checklist

Procedure: 1. Linked lesson: Level 2 Ch 6 2. (Driver) Navigate to a password strength checking website (e.g., howsecureismypassword.net). 3. (Navigator) Test a simple password like 'password123' and note the estimated crack time. 4. (Checker) Test a longer password with mixed characters like 'Sc#00l@2025!'. 5. (Driver) Test a passphrase like 'MyDogLovesToRunInPark2025'. 6. (Navigator) Compare the crack times for all three password types. 7. (Checker) Read the password security guidelines handout. 8. (Driver) Create three different strong passwords following the guidelines. 9. (Navigator) Evaluate each password using the strength checker. 10. (Checker) Discuss the trade-off between password strength and memorability. 11. (Driver) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Navigator) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. Why are short passwords with special characters weaker than long passphrases? 3. What makes a password easy to remember but hard to crack? 4. Why should you never reuse passwords across different accounts? 5. How often should you change your important passwords?

Activity 6.2: Privacy Settings Review

Activity 6.2: Privacy Settings Review

Duration: 40 minutes

Resources Needed:

- Computer with internet access
- Social media privacy settings guide
- Privacy audit checklist
- Sample privacy policy document

Procedure: 1. Linked lesson: Level 2 Ch 6 2. (Driver) Log in to a social media account (or review a sample account). 3. (Navigator) Navigate to the Privacy Settings section of the account. 4. (Checker) Review who can see your posts: Public, Friends, or Only Me. 5. (Driver) Check the profile visibility settings and adjust as needed. 6. (Navigator) Review the list of third-party apps connected to your account. 7. (Checker) Remove any apps you no longer use or recognise. 8. (Driver) Check the location sharing settings and disable if unnecessary. 9. (Navigator) Review the ad personalisation settings. 10. (Checker) Read a sample privacy policy and identify key data collection points. 11. (Driver) Document your changes and create a personal privacy checklist. 12. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 13. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between Public and Private account settings? 3. Why is it important to review third-party app permissions? 4. What information can social media platforms collect about you? 5. How can you protect your personal information online?

Dormitory Tasks

Task 1: Network Troubleshooting

Task 1: Network Troubleshooting

Duration: 30 minutes

Given a scenario where a computer cannot connect to the internet, list five possible causes and the steps you would take to diagnose and fix each one. Use the `ping` and `tracert` commands in your explanation.

Task 2: Email Professional Draft

Task 2: Email Professional Draft

Duration: 30 minutes

Write a formal email to the school principal requesting permission to organise an IT exhibition. Include a proper subject line, greeting, body with at least three paragraphs, closing, and your signature. Attach a proposed event plan as a separate document.

Task 3: GIMP Photo Restoration

Task 3: GIMP Photo Restoration

Duration: 30 minutes

Given a faded or low-quality image, use GIMP tools to improve its brightness, contrast, and sharpness. Add a watermark text layer with your name at 30% opacity. Save the edited image in XCF format with all layers intact.

Task 4: Audio Podcast Intro

Task 4: Audio Podcast Intro

Duration: 30 minutes

Using Audacity, create a 30-second podcast introduction. Include a voice recording, background music with fade-in and fade-out effects, and a sound effect transition. Export the final audio as an MP3 file.

Task 5: Presentation Design

Task 5: Presentation Design

Duration: 30 minutes

Create a 6-slide presentation on 'The History of the Internet'. Include a title slide, four content slides with images, and a conclusion slide. Add speaker notes for each content slide and apply consistent transitions.

Task 6: Data Analysis Report

Task 6: Data Analysis Report Duration: 30 minutes

Given a dataset of 15 students with marks in four subjects, create a spreadsheet that calculates total, average, highest, and lowest marks. Create a bar chart comparing student totals and a pie chart showing subject-wise distribution.

Task 7: Scratch Maze Game

Task 7: Scratch Maze Game Duration: 30 minutes

Design a simple maze game in Scratch where the player navigates a sprite through a maze to reach a goal. Use arrow key controls, wall collision detection, and a timer that stops when the goal is reached.

Task 8: Password Audit

Task 8: Password Audit Duration: 30 minutes

Create a report evaluating the strength of five different passwords. For each password, document its length, character types, and estimated crack time using an online tool. Provide recommendations for improving weak passwords.

Task 9: Spreadsheet Budget

Task 9: Spreadsheet Budget Duration: 30 minutes

Create a monthly budget spreadsheet for a school event with at least 10 expense categories. Use SUM for totals, calculate percentage of budget for each category, and create a pie chart showing expense distribution.

Task 10: Scratch Animation Story

Task 10: Scratch Animation Story Duration: 30 minutes

Create a 30-second animated story in Scratch with at least three sprites, background changes, dialogue using speech bubbles, and sound effects. Use broadcast messages to coordinate sprite actions.

Task 11: Image Collage

Task 11: Image Collage Duration: 30 minutes

Using GIMP, create a photo collage with at least five images. Use layers for each image, apply different blending modes, add text labels, and create a decorative border. Export as both XCF and PNG.

Task 12: Privacy Policy Summary

Task 12: Privacy Policy Summary Duration: 30 minutes

Read the privacy policy of a popular website or app. Write a one-page summary covering: what data is collected, how it is used, who it is shared with, and what choices users have. Highlight any concerns you identify.

Capstone Projects

Capstone Project 1: School Digital Newsletter

Activity CP1: School Digital Newsletter

Duration: 2 weeks

Resources Needed:

- Computer with internet access
- LibreOffice Writer and Impress
- GIMP for image editing
- Audacity for audio clips
- School network or USB drive for sharing

Procedure: 1. (Driver) Form a team of 4-5 students and assign roles: Editor, Writer, Designer, Photographer. 2. (Navigator) Plan the newsletter content: articles, event reports, interviews, and puzzles. 3. (Checker) Write and edit at least three articles using word processing features. 4. (Driver) Create images and graphics using GIMP for article illustrations. 5. (Navigator) Record a short audio introduction using Audacity. 6. (Checker) Design the newsletter layout using a desktop publishing tool or presentation software. 7. (Driver) Apply consistent formatting: fonts, colours, headings, and page numbers. 8. (Navigator) Review and proofread the entire newsletter as a team. 9. (Checker) Export the final newsletter as a PDF and share with the class. 10. (Driver) Present the newsletter creation process to the class. 11. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What role did each team member play in the project? 3. How did you ensure consistency in formatting across different articles? 4. What challenges did you face while collaborating and how did you overcome them? 5. How could you distribute this newsletter to the entire school electronically?

Capstone Project 2: Interactive Learning Game

Activity CP2: Interactive Learning Game

Duration: 2 weeks

Resources Needed:

- Computer with Scratch (scratch.mit.edu)
- Game design document template
- Art assets (sprites and backgrounds)
- Sound effects and music files
- Testing feedback form

Procedure: 1. (Driver) Choose a subject topic (Science, Math, or English) for your game. 2. (Navigator) Design the game concept: objective, rules, levels, and scoring system. 3. (Checker) Create a game design document with flowcharts and sprite descriptions. 4. (Driver) Program the main game mechanics using Scratch blocks. 5. (Navigator) Add at least three levels with increasing difficulty. 6. (Checker) Include score tracking, timer, and life system. 7. (Driver) Add visual and audio feedback for correct and incorrect answers. 8. (Navigator) Create a start screen, instructions screen, and game over screen. 9. (Checker) Test the game with classmates and collect feedback. 10. (Driver) Revise the game based on feedback and prepare a final presentation. 11. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. How does your game reinforce learning of the chosen subject? 3. What programming concepts did you use most frequently? 4. How did user feedback improve your game design? 5. What would you add or change if you had more time?

Computer Club Activities

Club Activity 1: Binary Number Challenge

Activity CA1: Binary Number Challenge

Duration: 40 minutes

Resources Needed:
- Binary conversion reference sheet
- Binary puzzle cards
- Calculator
- Binary game software or online tool

Procedure: 1. (Driver) Review the binary number system and how binary relates to decimal. 2. (Navigator) Convert 10 decimal numbers to binary using the division method. 3. (Checker) Convert 10 binary numbers to decimal using the positional value method. 4. (Driver) Complete a binary puzzle worksheet with 20 conversion problems. 5. (Navigator) Use an online binary game to test your speed and accuracy. 6. (Checker) Participate in a timed binary conversion race with classmates. 7. (Driver) Discuss how computers use binary to represent all data. 8. (Navigator) Explore binary representation of text using ASCII codes. 9. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 10. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. Why do computers use binary instead of decimal? 3. How many bits are needed to represent the number 100 in binary? 4. What is the relationship between binary and hexadecimal number systems? 5. How is the letter 'A' represented in binary using ASCII?

Club Activity 2: Cybersecurity Quiz

Activity CA2: Cybersecurity Quiz

Duration: 40 minutes

Resources Needed:
- Quiz questions on cybersecurity
- Buzzers or response cards

- Score sheet
- Cybersecurity awareness posters

Procedure: 1. (Driver) Divide the class into teams of 3-4 students. 2. (Navigator) Review cybersecurity concepts: phishing, malware, social engineering. 3. (Checker) Conduct a round of multiple-choice questions on password security. 4. (Driver) Play a scenario-based round where teams identify security threats. 5. (Navigator) Complete a true or false round on internet safety practices. 6. (Checker) Discuss the correct answers after each round. 7. (Driver) Calculate team scores and announce the winning team. 8. (Navigator) Create a cybersecurity awareness poster based on what you learned. 9. (Checker) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 10. (Driver) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is phishing and how can you identify a phishing email? 3. What should you do if you suspect your account has been compromised? 4. How does two-factor authentication improve account security? 5. What are the dangers of using public Wi-Fi networks?

Club Activity 3: Web Design Basics

Activity CA3: Web Design Basics

Duration: 40 minutes

Resources Needed:
- Computer with text editor (Notepad or VS Code)
- HTML and CSS reference sheets
- Sample HTML templates
- Web browser for preview

Procedure: 1. (Driver) Open a text editor and create a new file named index.html. 2. (Navigator) Write the basic HTML structure: html, head, body tags. 3. (Checker) Add a heading, paragraph, and an image using HTML tags. 4. (Driver) Create a simple navigation menu using an unordered list. 5. (Navigator) Add a table with at least 4 rows and 3 columns. 6. (Checker) Create a separate CSS file and link it to the HTML document. 7. (Driver) Apply styling: background colour, font family, and text alignment. 8. (Navigator) Style the navigation menu with hover effects. 9. (Checker) Open the HTML file in a web browser to preview the design. 10. (Driver) Validate the HTML using an online validator tool. 11. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What is the difference between HTML and CSS? 3. Why should you use semantic HTML tags like header, nav, and footer? 4. How does the browser interpret HTML and CSS files? 5. What is the purpose of the DOCTYPE declaration?

Club Activity 4: Spreadsheet Olympics

Activity CA4: Spreadsheet Olympics

Duration: 40 minutes

Resources Needed:
- Computer with LibreOffice Calc

- Competition datasets
- Formula reference cards
- Timer for timed challenges

Procedure: 1. (Driver) Divide the class into teams for the Spreadsheet Olympics. 2. (Navigator) Round 1: Speed Data Entry - enter 20 data rows as fast as possible. 3. (Checker) Round 2: Formula Challenge - solve 10 calculation problems using formulas. 4. (Driver) Round 3: Chart Creation - create the most informative chart from given data. 5. (Navigator) Round 4: Error Finding - identify and fix errors in a spreadsheet. 6. (Checker) Round 5: Pivot Table Puzzle - summarise data using pivot tables. 7. (Driver) Score each round based on accuracy and speed. 8. (Navigator) Discuss the most efficient approaches for each challenge. 9. (Checker) Announce the Spreadsheet Olympics champion team. 10. (Driver) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Navigator) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. Which formula challenge was the most difficult and why? 3. How did you decide which chart type to use in Round 3? 4. What strategies did you use to find errors quickly in Round 4? 5. How can spreadsheet skills be applied in real-world situations?

Club Activity 5: Scratch Game Jam

Activity CA5: Scratch Game Jam

Duration: 40 minutes

Resources Needed:
- Computer with Scratch (scratch.mit.edu)
- Game idea brainstorming worksheet
- Sprite and backdrop library
- Timer for the jam session

Procedure: 1. (Driver) Set a 30-minute timer for the Game Jam challenge. 2. (Navigator) Each student or pair chooses a game theme: platformer, puzzle, or shooter. 3. (Checker) Program the core game mechanic within the time limit. 4. (Driver) Add at least one enemy or obstacle with collision detection. 5. (Navigator) Include a scoring system and a game over condition. 6. (Checker) Add basic sound effects for key actions. 7. (Driver) When the timer ends, present your game to the group. 8. (Navigator) Vote on the most creative, most fun, and best-designed games. 9. (Checker) Discuss what you would add with more time. 10. (Driver) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 11. (Navigator) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. What was the most challenging part of creating a game in 30 minutes? 3. How did you prioritise features within the time limit? 4. What feedback did you receive from other players? 5. How would you improve your game if you had an extra hour?

Club Activity 6: Digital Art Competition

Activity CA6: Digital Art Competition

Duration: 40 minutes

Resources Needed:

- Computer with GIMP installed
- Drawing tablet (if available)
- Art theme prompt cards
- Judging criteria sheet

Procedure: 1. (Driver) Draw a theme card to determine the competition topic. 2. (Navigator) Plan your digital artwork on paper before starting in GIMP. 3. (Checker) Create a new canvas with appropriate dimensions. 4. (Driver) Use layers to build your artwork from background to foreground. 5. (Navigator) Apply at least three different GIMP tools: brush, gradient, and text. 6. (Checker) Use colour adjustments to enhance the final appearance. 7. (Driver) Add a signature or watermark layer. 8. (Navigator) Save the work in XCF format with layers and export as PNG. 9. (Checker) Present your artwork and explain your creative process. 10. (Driver) Vote on categories: Most Creative, Best Technique, and Most Detailed. 11. (Navigator) (Library station) Paper pack: plan or sketch this artefact on graph paper and predict results before your next lab turn. 12. (Checker) (All, last 4 min) Save work to shared drive, take screenshot, complete individual exit ticket.

Questions: 1. Predict (PRIMM): If you changed one input or setting here, what output change would you expect? Verify next session. 2. How did planning on paper help your digital artwork? 3. Which GIMP tool was most useful for your artwork and why? 4. How did working with layers simplify the creation process? 5. What would you do differently next time to improve your artwork?